ATTENDANCE RECOVERY COMPUTATION

Project Report

1. Problem Statement

Develop a Python program that calculates current attendance percentages and determines the minimum consecutive classes required to reach an eligibility threshold under different future attendance scenarios.

2. Objectives

• Calculate current attendance percentage.

• Check attendance against the eligibility threshold.

• Find minimum consecutive classes required to reach the threshold.

• Evaluate different future attendance rates.

• Display the final attendance percentage for each scenario.

3. Assumptions

• The student has attended 42 out of 60 classes.

• The eligibility threshold is 75%.

• Future attendance is represented by a rate such as 100%, 90%, or 80%.

• Classes are counted as whole classes.

4. Algorithm

1. Start.

2. Read present classes, total classes, threshold, and future attendance rate.

3. Calculate current attendance.

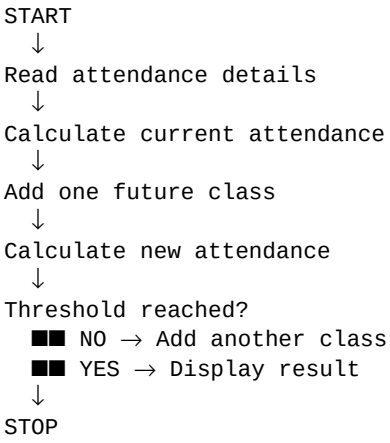
4. Add future classes one by one.

5. Recalculate attendance after each class.

6. Stop when the threshold is reached.

7. Display required classes and final attendance.

5. Flowchart


START
Read attendance details
Calculate current attendance
Add one future class
Calculate new attendance
Threshold reached?
NO → Add another class
YES → Display result
STOP

6. Pseudocode

```
BEGIN
INPUT present, total, threshold, future_rate
classes = 0
WHILE new attendance < threshold:
    classes = classes + 1
    calculate new attendance
DISPLAY current attendance
DISPLAY minimum classes
DISPLAY final attendance
END
```

7. Python Source Code

```
def recovery_classes(present, total, threshold, rate):
    classes = 0
    while ((present + rate * classes)
           / (total + classes) * 100 < threshold):
        classes += 1
    return classes

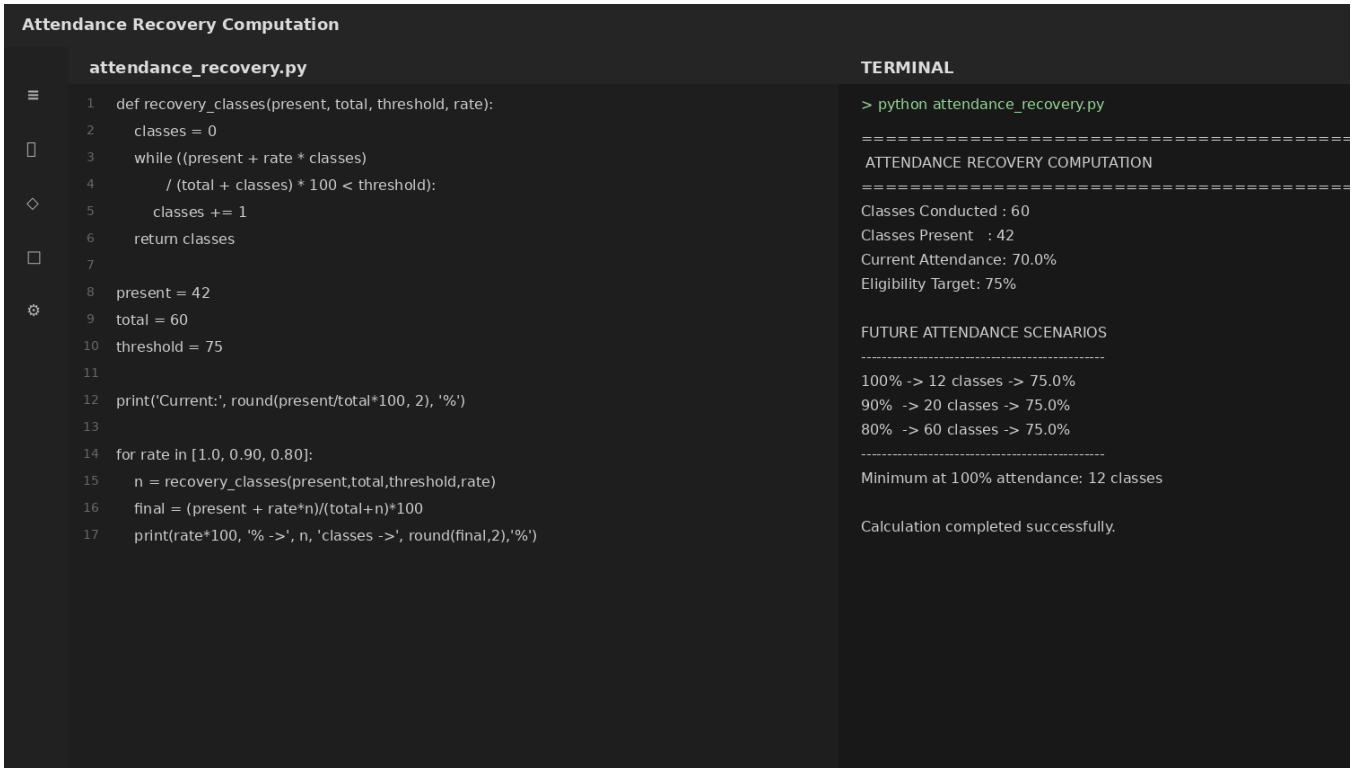
present = 42
total = 60
threshold = 75

print("Current:", round(present / total * 100, 2), "%")

for rate in [1.0, 0.90, 0.80]:
    n = recovery_classes(present, total, threshold, rate)
    final = (present + rate*n) / (total+n) * 100
    print(rate*100, "% ->", n, "classes ->", round(final, 2), "%")
```

8. Output Screenshot

VS Code-style source and terminal output screenshot.



9. Test Cases

Scenario	Future Attendance	Required Classes	Final Attendance
Perfect attendance	100%	12	75.00%
High attendance	90%	20	75.00%
Moderate attendance	80%	60	75.00%

10. Conclusion

The program calculates current attendance and determines the minimum number of future classes needed to reach the required eligibility threshold. It also compares different future attendance scenarios.